ENERGETICS OF MUSCLE CONTRACTION AND OXYGEN DEBT

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LEARNING OBJECTIVES:

- Define ATP as immediate source of energy for muscle contraction
- Explain how ATP is regenerated during mucle activity
- Describe the three energy systems that supply during muscle activity
- Identify the duration of activity supported by each system
- Identify which energy system predominates during short high-intensity exercise & prolonged endurance exercise
- Define EPOC(oxygen debt)
- Explain why oxygen level remain elevated after exercise
- Distinguish b/w rapid and slow component of EPOC
- Identify the factors affecting EPOC

INTRODUCTION:

Skeletal muscles contraction and relaxation requires a continuous supply of energy.When a muscle contracts against a load, it performs work.To perform work means the energy is transferred from the muscle to the external load to lift an object to a greater height or to overcome resistance to movement.

In mathematical terms, work is:

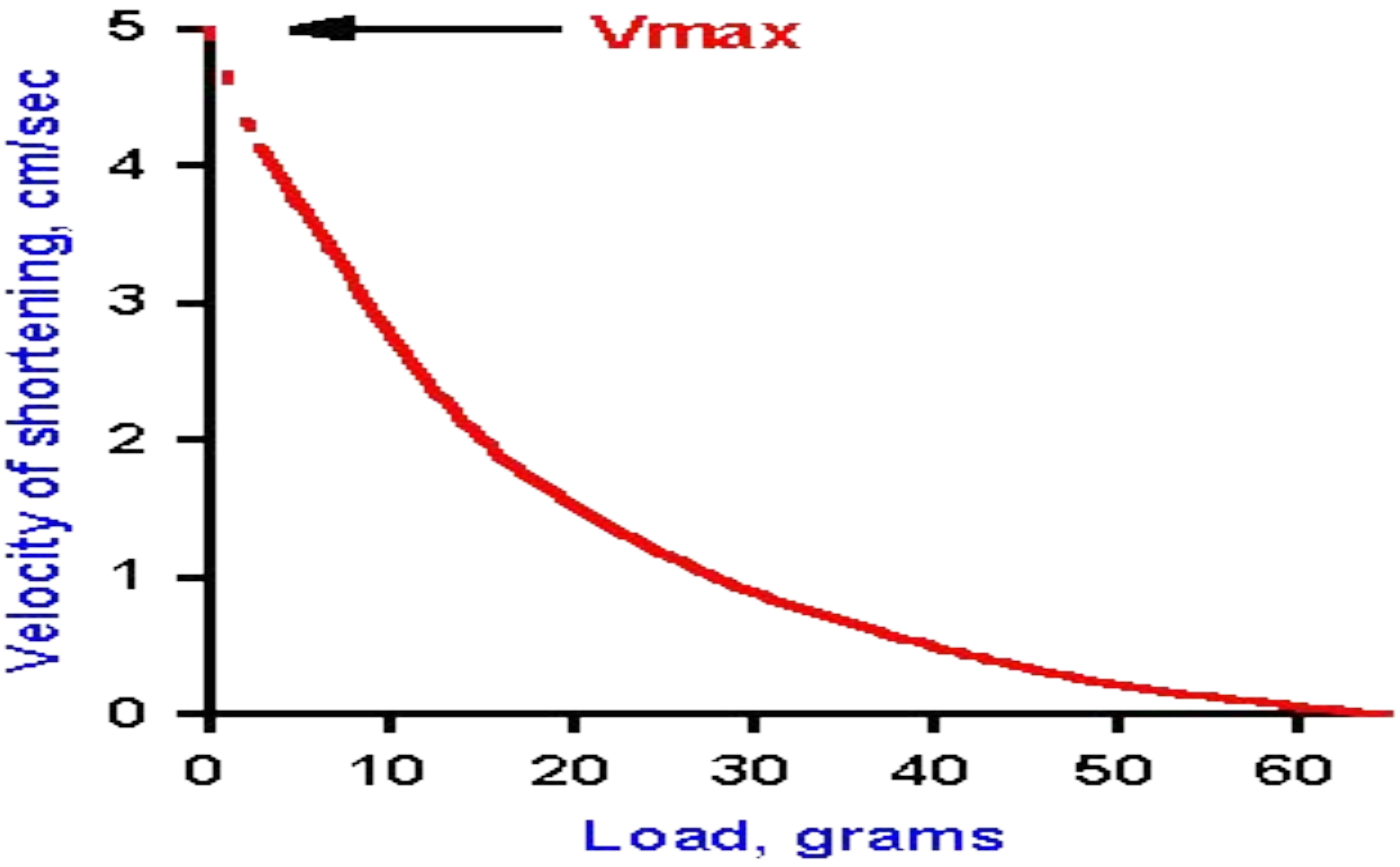
W=L (multiply) D

Where,

W= work output

L= load

D= distance of movement against the load



Vmax
Velocity of shortening, cm/sec
5
4
3
2
1
0
0
10
20
30
40
50
60
Load, grams

WHY NEED ENERGY?

Most of the energy required for muscle contraction is used to trigger the walk-along mechanism but small amounts of energy are required for:

1. Pumping of Ca^{+2} from sarcoplasm into sarcoplasmic reticulum after contraction is over.
2. Pumping Na^{+} & K^{+} through the muscle fibre membrane to maintain appropriate ionic environment for the propagation of muscle fibre action potential.

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This energy is derived from ATP. Stored ATP in the muscles (about 4 millimolar) lasts only for 1-2 seconds. So continuous regeneration of ATP is required for the muscles to work. Failure of energy supply leads to fatigue and weakness.

ENERGY SYSTEMS:

There are three major energy systems:

1. Phosphagen system(creatine phosphate)
2. Glycolytic system
3. Oxidative phosphorylation

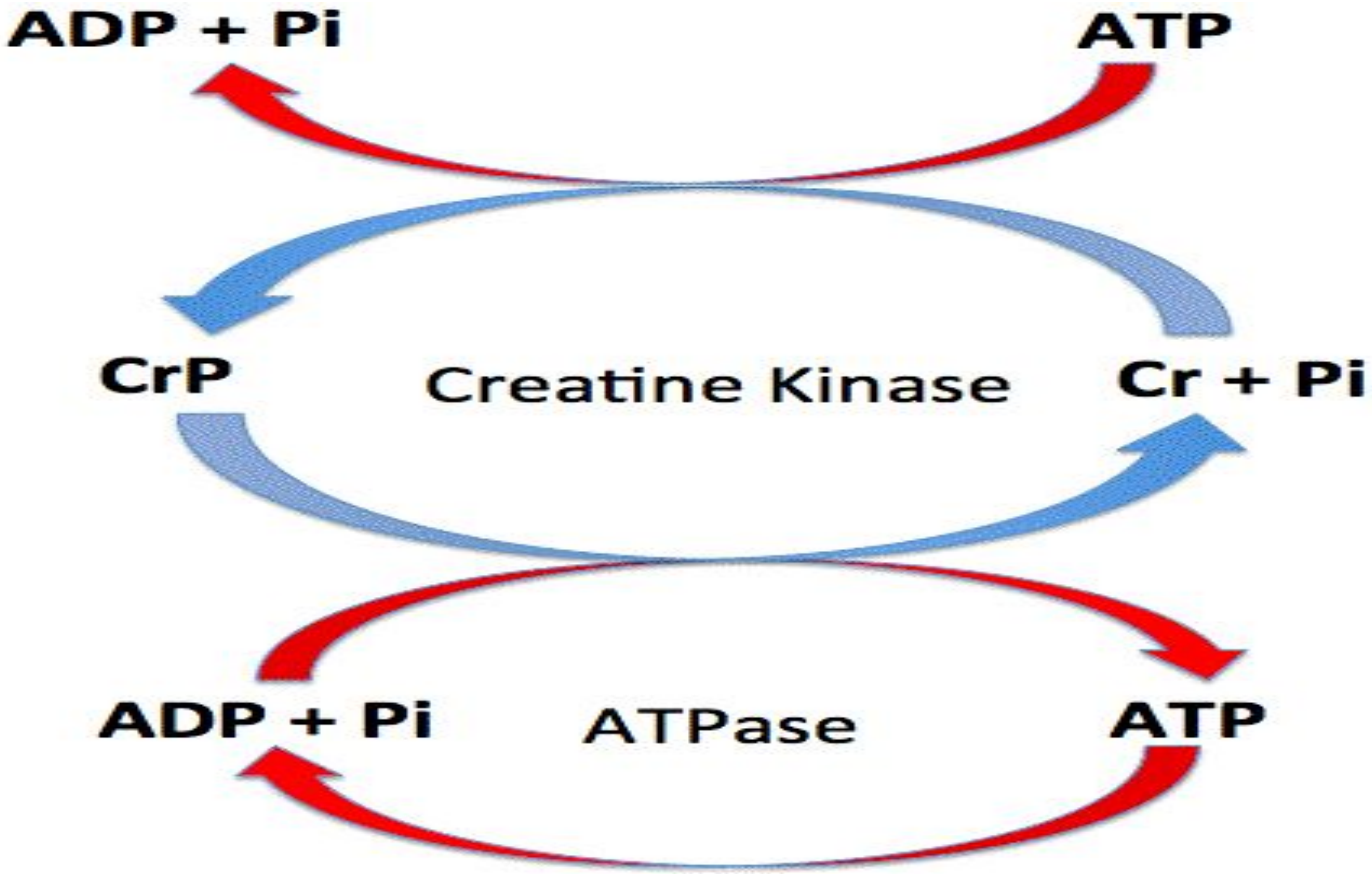
1. PHOSPHAGEN SYSTEM:

The first source of energy that is used to reconstitute the ATP is the substance ***phosphocreatine***, which carries a high energy phosphate bond similar to the bonds of ATP. Therefore phosphocreatine is instantly cleaved and its released energy causes bonding of a new phosphate ion to ADP to reconstitute the ATP.

However, the total amount of phosphocreatine in the muscle fibre is also small, only about 5 times as great as the ATP. Therefore the combined energy of both the stored ATP and the phosphocreatine in the muscle is capable of causing maximal muscle contraction for only 5-8 seconds.

Characteristics:

1) Very rapid ATP production
2) No oxygen required
3) No lactic acid production
4) Limited capacity(exhausted in 10 sec)
5) Weight lifting, jumping


ADP + Pi
ATP
CrP
Creatine Kinase
Cr + Pi
ADP + Pi
ATPase
ATP

2. GLYCOLYTIC SYSTEM(GLYCOLYSIS):

The second important source of energy which is used to reconstitute both ATP and phosphocreatine is a process called ***glycolysis***(the breakdown of glycogen previously stored in muscle cells. Rapid enzymatic breakdown of glycogen to pyruvic acid and lactic acid liberates energy that is used to convert ADP to ATP. The ATP can then be used directly to energize additional muscle contraction and also to reform the stores of phosphocreatine.

The importance of this glycolysis mechanism is twofold. First, glycolytic reactions can occur even in the absence of oxygen, so muscle contraction can be sustained for many seconds and sometimes upto more than 1minute, even when oxygen delivery from the blood is not available.

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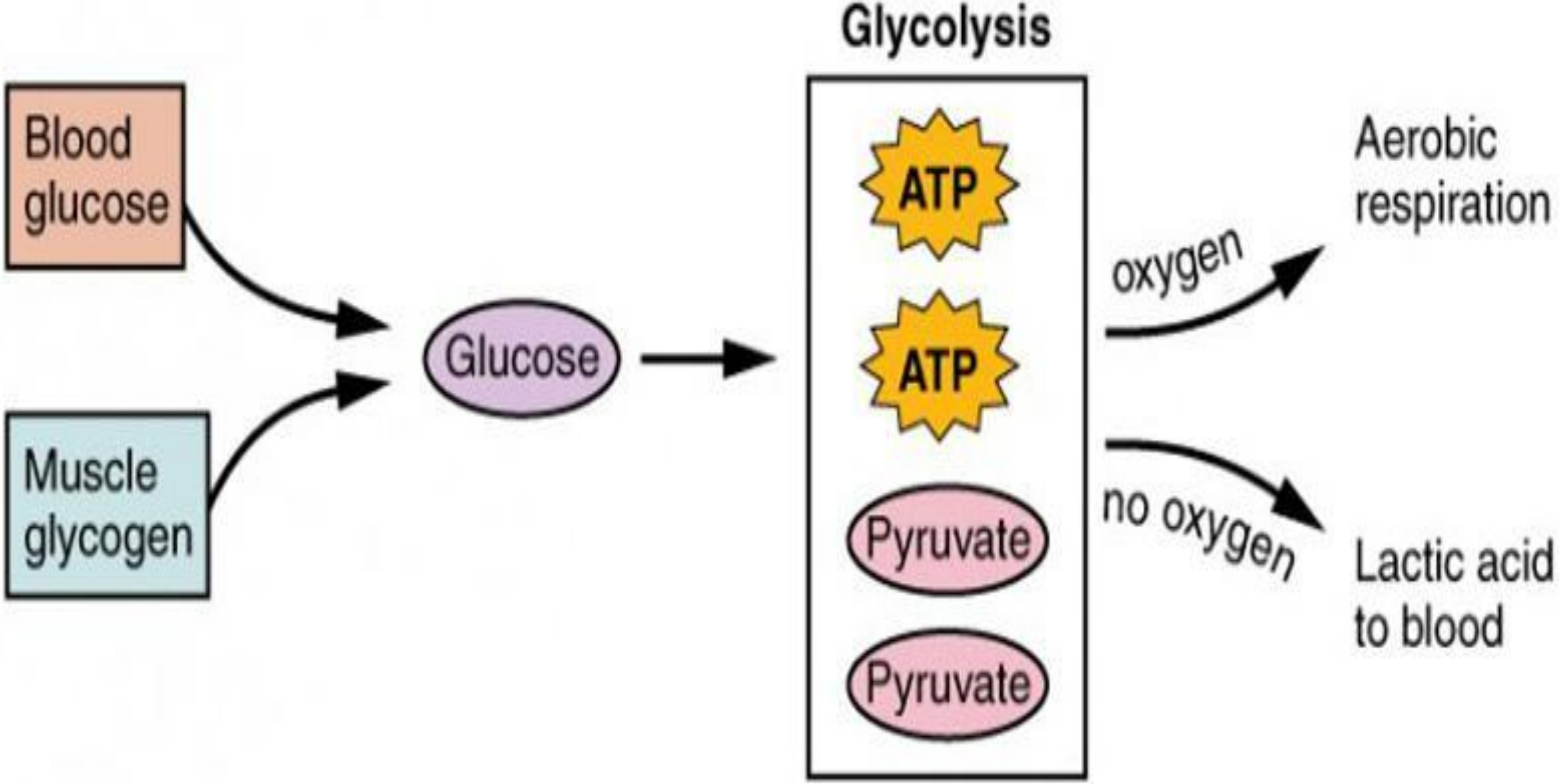
Second, the rate of ATP formation by glycolysis is about 2.5 times as rapid as ATP formation in response to cellular foodstuffs reacting with oxygen. However, this system loses its capability to sustain maximum muscle contraction after about 1 minute.

Characteristics:

1) Fast ATP production
2) No oxygen required
3) Produces lactate
4) Causes decrease pH(acidosis)
5) Supports activity for 30 sec – 2min
6) 400 – 800m run(short period running)

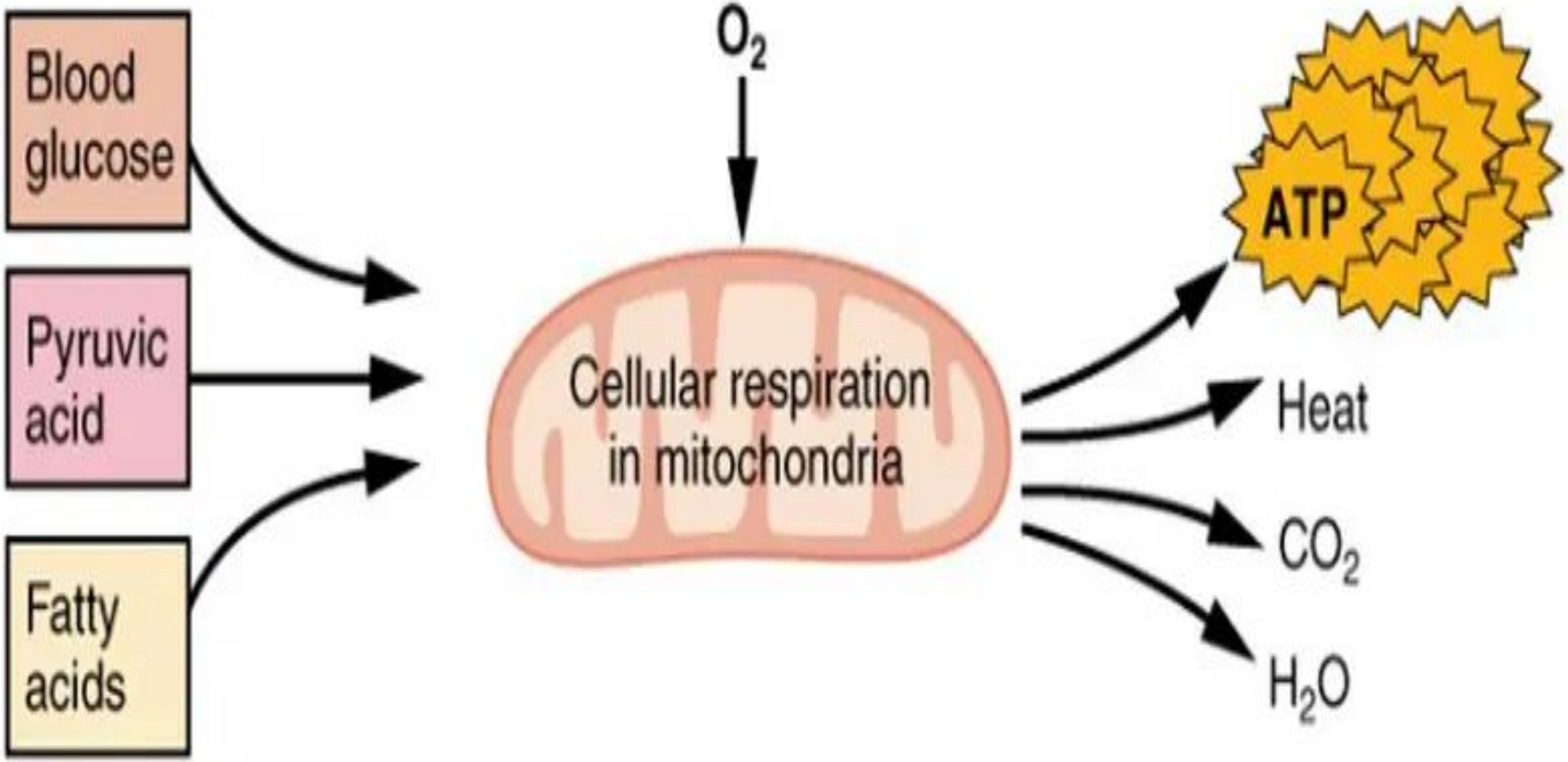

Glycolysis
Blood glucose
Muscle glycogen
Glucose
ATP
ATP
Pyruvate
Pyruvate
oxygen
Aerobic respiration
no oxygen
Lactic acid to blood

3. OXIDATIVE METABOLISM:

The 3rd and final source of energy is ***oxidative metabolism***, which which means combining oxygen with the end products of glycolysis and with various other cellular foodstuffs to liberate ATP. The foodstuffs that are consumed are carbohydrates, fats and proteins. For extremely long-term maximal muscle activity(over many hrs) the greatest proportion of energy comes from fats but for periods of 2-4 hrs as much as one half of energy can come from stored carbohydrates.

Characteristics:

1) Slow ATP production
2) Very high capacity
3) No lactate accumulation
4) Supports long-duration activity
5) Marathon, cycling, postural muscle activity


Blood glucose
Pyruvic acid
Fatty acids
O_2
Cellular respiration in mitochondria
ATP
Heat
CO_2
H_2O

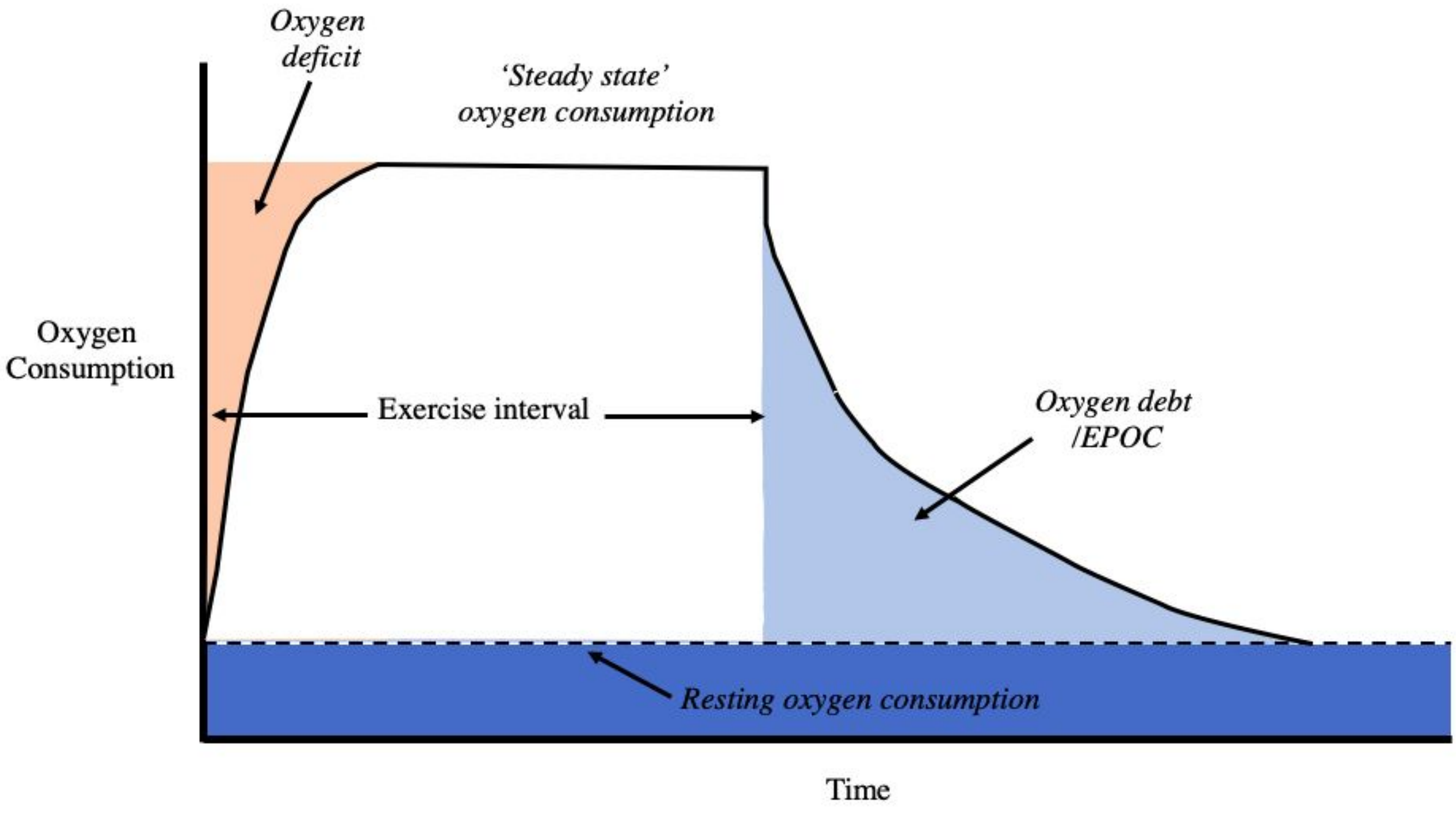
ENERGY SYSTEM CONTRIBUTION OVER TIME

- 0 – 10 SEC -> PHOSPHAGEN SYSTEM
- 10 SEC - 2 MIN-> ANAEROBIC GLYCOLYSIS
- > 2 MIN -> AEROBIC RESPIRATION

EXCESS POST-EXERCISE OXYGEN CONSUMPTION(OXYGEN DEBT)

After exercise extra oxygen is required by the body to restore the normal physiology. Oxygen debt can be defined as:

The extra oxygen required after exercise to restore the body to its pre-exercise state.


Oxygen deficit
'Steady state' oxygen consumption
Oxygen Consumption
Exercise interval
Oxygen debt /EPOC
Resting oxygen consumption
Time

Why does oxygen debt occurs?

During intense exercise:

1) ATP and creatine phosphate are depleted
2) Accumulation of lactate
3) Myoglobin becomes deoxygenated
4) Body temperature increases
5) Heart rate and ventilation increases

Why extra oxygen is needed after exercise?

After exercise extra oxygen is required to:

1) Replenish ATP stores
2) Reoxygenate myoglobin
3) Convert lactate to glucose again in the liver(cori cycle)
4) Oxidize lactate to CO_2 AND H_2O
5) Normalize body temperature to its pre-exercise state
6) Restore hormonal balance

COMPONENTS OF EPOC

- Fast component (alactacid phase)
- Slow component (lactacid phase)

Fast component:

It lasts for about 2-3 min after exercise.During this phase ATP is replenished, creatine phosphate is restored and deoxygenated myoglobin is reoxygenated again.

Slow component:

This phase lasts for 30 min to several hours. During this phase lactate is converted back to pyruvate and then into glocuse and this glucose is transported back to muscles.Lactate metabolism also occur by heart and skeletal muscles to form carbondioxide and water.

FACTORS AFFECTING EPOC:

The magnitude of EPOC depends upon:

1) Intensity of exercise
2) Duration of exercise
3) Muscle fibre type
4) Contribution of anaerobic energy supply during exercise
5) Body temperature
6) Fitness level

TYPES OF MUSCLE FIBRES

1. TYPE I MUSCLE FIBRES:

These are the slow twitch muscle fibres also called ***slow-oxidative fibres*** or ***red muscle fibres***. They are characterized by slow contraction speed, low myosin-ATPase activity. They are highly resistant to fatigue and have high no. of mitochondria and myoglobin content. Type I fibres have rich capillary supply due to which these fibres are red in colour.

Metabolism: Primarily aerobic respiration occur in these fibres.High oxidative phosphorylation causes efficient ATP production. Fatty acids are commonly used as fuel in these fibres. Type I fibres are well suited for prolonged activity.

Example: Postural muscles, marathon runners.

2. TYPE II MUSCLE FIBRES:

Type II muscle fibres are fast twitch muscle fibres also called ***white fibres***(especially II b).These fibres show rapid contraction and have high myosin-ATPase activity.These fibres exhibit faster cross-bridge cycling and gives higher force output.

Type II(a) fibres: These are the fast oxidative glycolytic fibres and show intermediate fatigue resistance. Both aerobic and anaerobic type of metabolism occurs in these fibres. These fibres exhibit moderate no. of mitochondria and seen in middle-distance athletes.

Type II(b) fibres: These fibres show very rapid contraction and gives high work output. These fibres are characterized by low mitochondrial density and fatigue quickly. Type II(b) fibres typically depend upon anaerobic glycolysis.

Features	Type 1 (slow twitch) fibers	Type 2 (fast twitch) fibers
Mitochondria	High	Low
Myoglobin	High	Low
Capillary density	High	Low
Energy source	Aerobic	**Anaerobic** (with some aerobic mixed in for type 2a fibers)
Contraction speed	Slow	Fast
Fatigue resistance	High	Low
Primary function	Endurance	Short, explosive power
Color	Red	**White** (or pink for type 2a)

POST ASSESSMENT

Q1. What is the immediate source of energy for muscle contraction?

a) Glucose
b) ATP
c) Glycogen
d) Lactic acid

Q2. During high intensity exercise which process produces ATP without oxygen?

a) Oxidative phosphorylation
b) Anaerobic glycolysis
c) Krebs cycle
d) Electron transport chain

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Q3. Why does oxygen consumption remain elevated after exercise

a) To remove carbondioxide
b) To increase heart rate
c) To build new muscle
d) To restore the body to resting state

Q4. ATP is responsible for both muscle contraction and relaxation(T/F)

Q5. Lactic acid accumulation causes muscle fatigue and weakness(T/F)

Q6. Anaerobic glycolysis occurs in mitochondria(T/F)

THANK YOU